

Two-photon Physics at a lepton-hadron collider at CERN

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(on behalf of HELP4CERN Study Group)

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Why a muon-hadron collider for HET physics?



- A muon-hadron collider combines the clean lepton-hadron initial state of DIS with a multi-TeV centre-of-mass energy, opening a new regime for Higgs, electroweak, top, and BSM studies.
- Relative to an electron-proton collider, replacing the electron beam by a muon beam strongly relaxes the synchrotron-radiation limitation, allowing significantly higher beam energies and hence much larger hard-scattering rates, and pushes the centre-of-mass energy deep into the multi-TeV regime.
- Relative to proton-proton collisions, the μp environment remains experimentally cleaner, with smaller QCD backgrounds for many precision and BSM channels.

Three complementary physics paths at LH μ C

LH μ C combines three powerful and complementary directions: a strong QCD programme, high-precision Higgs/top/electroweak measurements, and a powerful probe of physics beyond the Standard Model.

Benchmark working points used in this talk



- The present talk focuses on the physics case of the LH μ C, using the current working benchmarks:
- The first option: $E_\mu = 500$ GeV, $E_p = 7$ TeV, $\sqrt{s} \simeq 3.7$ TeV, with a $\mathcal{L}_{\text{int}} = 250 \text{ fb}^{-1}$.
- **The nominal baseline:** $E_\mu = 1$ TeV, $E_p = 7$ TeV, $\sqrt{s} \simeq 5.3$ TeV, with a $\mathcal{L}_{\text{int}} = 1000 \text{ fb}^{-1}$.
- The second option: $E_\mu = 2$ TeV, $E_p = 7$ TeV, $\sqrt{s} \simeq 7.5$ TeV, with a $\mathcal{L}_{\text{int}} = 1000 \text{ fb}^{-1}$.

LH μ C physics message

These working points are sufficient to demonstrate the main message of LH μ C: already at the 500 GeV, LH μ C enters the multi-TeV regime and becomes highly competitive for Higgs, top, electroweak, $\gamma\gamma$, and representative BSM studies.

From HELP4CERN White Paper (in preparation)

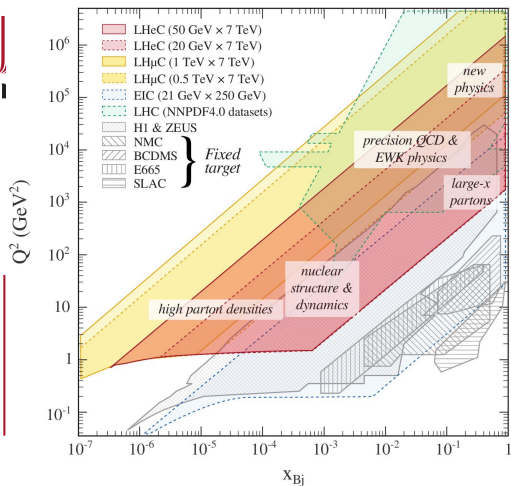
Benchmarks

Machine	Benchmark
LHeC	$E_e = 20, 50$ GeV
LH μ C (option 1)	$E_\mu = 500$ GeV
LH μ C (baseline)	$E_\mu = 1000$ GeV
LH μ C (option 2)	$E_\mu = 2000$ GeV
Configuration	Value
$\sqrt{s}_{\mu p}$ (option 1)	$\simeq 3.7$ TeV
$\sqrt{s}_{\mu p}$ (baseline)	$\simeq 5.3$ TeV
$\sqrt{s}_{\mu p}$ (option 2)	$\simeq 7.5$ TeV
$\mathcal{L}_{\text{int}}$ (option 1)	250 fb^{-1}
$\mathcal{L}_{\text{int}}$ (baseline)	1000 fb^{-1}
$\mathcal{L}_{\text{int}}$ (option 2)	1000 fb^{-1}



See talk by Krzysztof Piotrzkowski

Kinematic reach and QCD physics landscape



Available (x, Q^2) plane for the proposed LH μ C, compared with fixed-target, collider, and EIC kinematic coverage.

QCD program at LH μ C

- The multi-TeV μp configuration extends the DIS kinematic coverage well beyond HERA and significantly beyond the LHeC.
- The same machine that reaches extreme x and Q^2 also strengthens the QCD programme through improved access to proton PDFs, and precise determinations of α_s .
- This motivates the broader CERN bridge-project narrative: LH μ C strengthens both the QCD frontier and the HET/BSM programme.

See talk by Piotr Kotko

Representative HET benchmark cross sections

What does this table show?

LH μ C as a genuine HET factory: Higgs production is large enough to support precision Higgs physics, including benchmark coupling studies & the top sector benefits the most, enabling both precision top measurements and sensitivity to rare top processes, & even the 500 GeV option opens a qualitatively new regime for electroweak, EFT, and BSM physics.

From HELP4CERN White Paper (in preparation)

Benchmark inclusive cross sections

Process	20 GeV LHeC	50 GeV LHeC	500 GeV LH μ C	1 TeV LH μ C
$\sigma(\ell p \rightarrow \nu_\ell H + \text{jet} + X)$ [fb]	52	164	438	731
$\sigma(\ell p \rightarrow \ell H + \text{jet} + X)$ [fb]	5.6	19	135	221
$\sigma(\ell p \rightarrow \ell t\bar{t} + X)$ [pb]	0.0015	0.0323	1.03	2.23
$\sigma(\ell p \rightarrow \nu_\ell t + X)$ [pb]	0.55	2.89	21.26	36.77

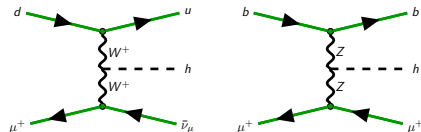
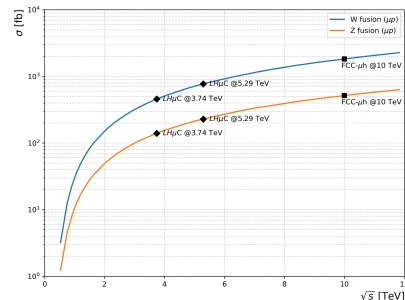
Key comparisons

Relative to the 50 GeV LHeC benchmark, the nominal 1 TeV LH μ C baseline enhances the benchmark inclusive rates by approximately: ~ 4.5 for charged-current Higgs production, ~ 12 for neutral-current Higgs production, ~ 13 for single-top production, and ~ 69 for $t\bar{t}$ production.

Higgs production at LH μ C: channels, rates, and statistics



- Higgs production is dominated by CC VBF, with a subleading but still sizeable NC contribution.
- Using the current benchmarks:
 - 500 GeV LH μ C: $\sigma_{CC} \simeq 438$ fb, $\sigma_{NC} \simeq 135$ fb,
 - 1 TeV LH μ C: $\sigma_{CC} \simeq 731$ fb, $\sigma_{NC} \simeq 221$ fb.
- For the benchmark integrated luminosities, these correspond to roughly:
 - 1.10×10^5 CC Higgs and 3.38×10^4 NC Higgs events at 500 GeV,
 - 7.31×10^5 CC Higgs and 2.21×10^5 NC Higgs events at 1 TeV.
- This already motivates precision Higgs-coupling studies beyond a mere rescaling of LHeC expectations.

CC: $\mu^+ p \rightarrow \bar{\nu}_\mu j h$ NC: $\mu^+ p \rightarrow \mu^+ j h$ 

Higgs boson production cross sections at LH μ C for the CC (W-fusion) and subleading NC (Z-fusion) contributions.

Precision Higgs case study: $h \rightarrow b\bar{b}$ and coupling prospects



- A concrete benchmark study already exists for the CC VBF channel $p\mu^+ \rightarrow j\bar{\nu}_\mu h; h \rightarrow b\bar{b}$.
- Analysis chain used in this study:
 - MG5_aMC@NLO + Pythia6 + FastJet + Delphes,
 - LHeC-like asymmetric detector setup and 75% b -tagging efficiency,
 - selection with exactly two b -jets and

$$|m_{bb} - m_h| < 25 \text{ GeV}.$$

- The statistical uncertainty estimated as $\frac{\Delta\sigma}{\sigma} = \frac{\sqrt{N_s + N_b}}{N_s}$,
- This gives a benchmark on the measurable rate which can be translated into a sensitivity on g_{hbb} :

$$\sigma(p\mu^+ \rightarrow j\bar{\nu}_\mu b\bar{b}) \propto \frac{g_{hWW}^2 g_{hbb}^2}{\Gamma_h}.$$

	LHμC-500	LHμC-1000
σ_{sig} [pb]	0.24	0.39
$\sigma_{\text{bgd-Z}}$ [pb]	0.30	0.50
$\sigma_{\text{bgd-mj}}$ [pb]	357.9	563.8
$\Delta\sigma/\sigma$	1.19%	0.69%
$\Delta g_{hbb}/g_{hbb}$	0.77%	0.60%

Benchmark CC $h \rightarrow b\bar{b}$ sensitivities. The coupling reach is obtained from the rate measurement after propagating the external uncertainty on g_{hWW}^2/Γ_h .

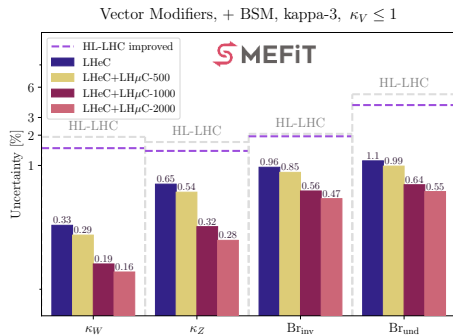
Key finding

The current LHμC benchmark study already reaches sub-percent sensitivity to g_{hbb} in the CC Higgs channel, improving on the LHeC expectation of about 0.97% [M. Tanaka, DIS2017] and substantially exceeding the HL-LHC expectation of about 4% [1902.00134 [hep-ph]].

Broader Higgs-coupling reach: a global-fit picture



- Beyond the dedicated $h \rightarrow b\bar{b}$ benchmark, we also examined a Higgs coupling fit in the kappa-3 framework [[JHEP 01 \(2020\) 139](#)].
 - These results are obtained in a first explorative study:
 - the same systematic factors as in the underlying LHeC-based setup,
 - the statistical error is rescaled using the benchmark LH μ C cross sections,
 - and the fit results shown here were obtained with the [SMEFIT framework](#) [[2604.16596](#)].
 - The strongest visible improvement is in the vector sector:
- $\delta\kappa_W \simeq 0.29\%$ (LH μ C-0.5TeV), 0.19% (LH μ C-1TeV).
- The clearest message of the current exploratory fit is the strong projected improvement in the vector sector, especially for κ_W and κ_Z , at the higher-energy LH μ C stages.

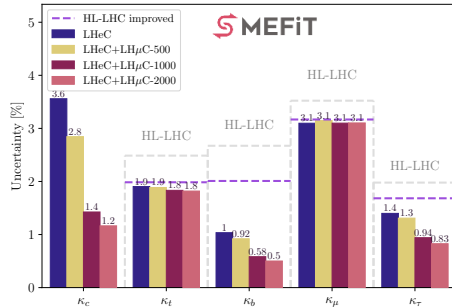
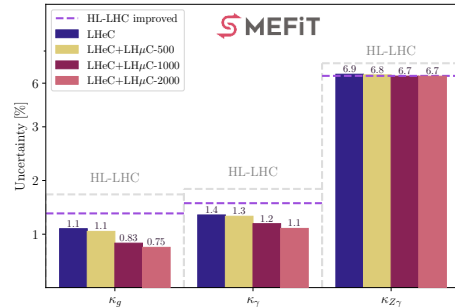


Exploratory vector-modifier result in the kappa-3 framework. The panel compares HL-LHC-based benchmarks with LHeC and with LHeC+LH μ C stages.

From HELP4CERN White Paper (in preparation)

Higgs&BSM@LHeC: Néstor Armesto

Broader Higgs-coupling reach: fermion and loop sectors

Fermion Modifiers, kappa-3, $\kappa_V \leq 1$ Loop Modifiers, kappa-3, $\kappa_V \leq 1$ 

- Fermion sector: the LH μ C can substantially improve the precision on selected fermion couplings, especially κ_b , κ_c , and κ_τ . Already at the 1 TeV stage, the combined LHeC+LH μ C fit becomes clearly stronger than the HL-LHC benchmark for κ_b and κ_c .
- Loop sector: loop-induced modifiers are not the primary strength of a VBF-driven Higgs programme. Nevertheless, the 1 TeV and 2 TeV stages make κ_g competitive with, and eventually better than, the HL-LHC benchmark.

Electroweak programme and EFT opportunities



EW precision reach at LH μ C

- Beyond Higgs rates, LH μ C offers a broader electroweak programme based on high- Q^2 DIS, vector-boson fusion, and photon-induced reactions.
- The larger kinematic span relative to the LHeC is expected to strengthen sensitivity to key electroweak observables, in particular m_W and $\sin^2 \theta_W$, while also extending the SMEFT lever arm.
- Overall, LH μ C is expected to extend the precision electroweak reach beyond that projected for the LHeC. **HET@LHeC: Paul Newman**

Highlighted EW / EFT opportunities

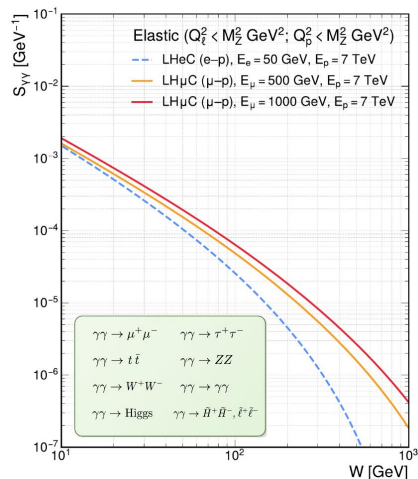
- High-scale measurements of the weak mixing angle and its running,
- Precision light-quark weak neutral-current couplings via γZ interference and Z exchange,
- Improved sensitivity to m_W in a clean DIS environment,
- Electroweak/top measurements linked to the Wtb vertex and direct sensitivity to V_{ts} ,
- SMEFT directions complementary to those probed at pp and e^+e^- machines,
- Photon-induced EFT sensitivity in $\gamma\gamma \rightarrow W^+W^-, ZZ$, and heavy final states.

High-energy $\gamma\gamma$ physics is a natural part of the LH μ C programme



Interesting HET directions

- LH μ C naturally operates as a *high-energy $\gamma\gamma$ collider*.
- In the nominal 5.3 TeV baseline, the effective $\gamma\gamma$ centre-of-mass energy extends into the multi-TeV regime, opening a broad programme of clean exclusive measurements.
 - anomalous τ moments,
 - $\gamma\gamma \rightarrow W^+W^-$ and anomalous quartic gauge couplings (aQGC),
 - loop-induced $\gamma\gamma \rightarrow ZZ$,
 - exclusive $t\bar{t}$ production,
 - light-by-light scattering (LbL)
 - BSM charged-particle pair production.

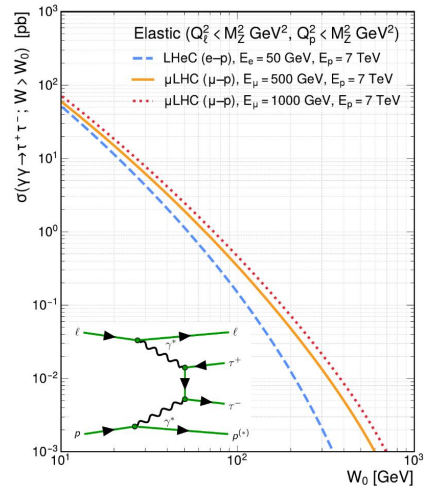


Elastic photon-photon luminosity spectrum $S_{\gamma\gamma}$ as a function of the $\gamma\gamma$ invariant mass W , comparing the LHeC and LH μ C benchmark configurations.

Benchmark $\gamma\gamma$ channel: $\tau^+\tau^-$ and anomalous τ moments

LH μ C $\tau^+\tau^-$ physics message

- The elastic $\gamma\gamma \rightarrow \tau^+\tau^-$ rate at LH μ C for $W_0 = 10$ GeV is already very large, with $\sigma_{\text{el}} \simeq 58.5$ pb at 3.7 TeV, and about 70.3 pb at the nominal 5.3 TeV baseline.
- This provides a high-statistics probe of the $\gamma\tau\tau$ vertex and anomalous magnetic (a_τ) and electric (d_τ) dipole moments.
- Anomalous contributions harden the high- $M_{\tau\tau}$ tail, making the invariant-mass spectrum the key discriminating observable.
- This is one of the cleanest examples of how LH μ C combines high energy with a low-background exclusive topology.

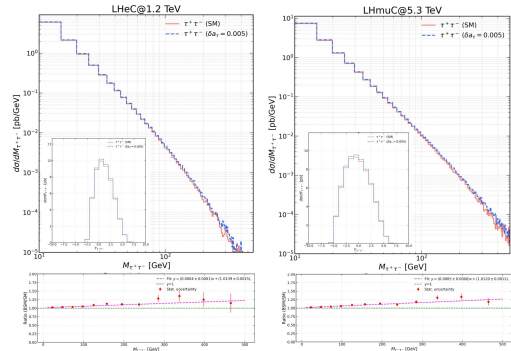


Elastic $\gamma\gamma \rightarrow \tau^+\tau^-$ cross section above the threshold W_0 , comparing the LHeC and LH μ C benchmark configurations.

Current limits and high- $M_{\tau\tau}$ sensitivity to δa_τ



- For the benchmark deformation $\delta a_\tau = 0.005$, the deviation from the SM remains small at low $M_{\tau\tau}$, but grows in the high- $M_{\tau\tau}$ tail; this is the main source of sensitivity.
- Compared with the LHeC benchmark, the nominal LH μ C setup probes a harder kinematic regime and retains better statistical reach in the tail, giving stronger sensitivity leverage on anomalous τ dipole couplings.
- These distributions therefore indicate good expected sensitivity at LH μ C, but a quantitative projected limit still requires a dedicated detector-level statistical analysis.



Tail-driven sensitivity

The sensitivity to anomalous τ dipole couplings is tail-driven. Relative to the LHeC benchmark, LH μ C combines a larger event sample with a stronger high- $M_{\tau\tau}$ lever arm, making it a promising benchmark for a_τ and d_τ studies.

Current LHC limits on a_τ

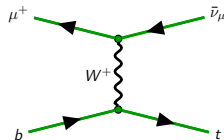
CMS exclusive pp photon-induced $\tau^+\tau^-$ production gives $a_\tau \in [-0.0022, 0.0041]$ [2406.03975 [hep-ex]], while the ATLAS high-mass $\tau^+\tau^-$ analysis gives $a_\tau \in [-0.0024, 0.0047]$ (95% CL) [2503.19836 [hep-ex]].

Top physics at LHμC: production mechanisms



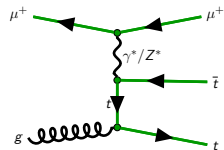
- The top sector provides one of the clearest HET benchmarks for LHμC, since both single-top and $t\bar{t}$ production become statistically powerful channels at the nominal baseline.
- **Single-top production** is dominated by charged-current DIS, with representative subprocess $\mu^+ b \rightarrow \bar{\nu}_\mu t$, mediated by t -channel W^+ exchange in the 5-flavor scheme.
- **Top-pair production** is dominated by neutral-current lepton–gluon scattering, with representative process $\mu^+ g \rightarrow \mu^+ t\bar{t}$, driven mainly by $\gamma^* g \rightarrow t\bar{t}$, while $Z^* g \rightarrow t\bar{t}$ is subleading.
- The top advantage is not only a formal cross-section enhancement: it is also supported by favourable acceptance, making the programme experimentally promising after realistic selections.

LO CC single-top production



CC DIS with t -channel W^+ exchange (5F scheme) with inclusive rates: $\sigma = 21.26$ pb, and 36.77 pb.

LO NC top-pair production



Dominant hard subprocess: $\gamma^* g \rightarrow t\bar{t}$; $Z^* g \rightarrow t\bar{t}$ is subleading with inclusive rates: $\sigma = 1.03$ pb, and 2.23 pb.

Why LH μ C is also a distinctive BSM machine



- $\mu^+ p$ collisions provide a distinctive BSM environment: a point-like muon beam scatters from quarks and gluons at multi-TeV centre-of-mass energy.
- The key advantage is channel-specific: μq and μg initial states can directly probe new particles coupled to second-generation leptons,

$$\mu q \rightarrow X, \quad \mu g \rightarrow X.$$

- This enables resonant single production of states that may be harder to isolate through pair production or inclusive searches at hadron colliders.
- The LH μ C is naturally powerful for BSM scenarios tied to second-generation leptons.
- Representative examples include:
 - R-parity-violating squark searches,
 - color-octet muons (leptogluons),
 - excited leptons,
 - contact interactions,
 - and other muon-coupled new-physics scenarios.

RPV squarks at LH μ C: resonant μq production



- R-parity-violating SUSY allows lepton-quark-squark couplings. The relevant superpotential terms are

$$W_{\text{RPV}} = \epsilon_i L_i \cdot H_u + \frac{1}{2} \lambda_{ijk} L_i \cdot L_j \bar{E}_k + \lambda'_{ijk} L_i \cdot Q_j \bar{D}_k + \frac{1}{2} \lambda''_{ijk} \bar{U}_i \bar{D}_j \bar{D}_k$$

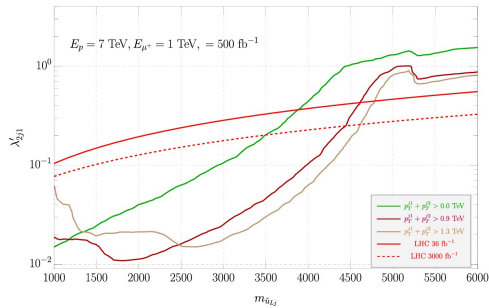
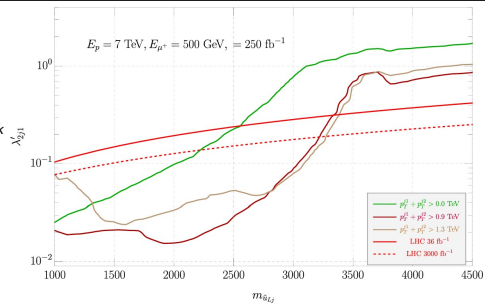
- The LH μ C benchmark focuses on the LQD -type term

$$W_{\text{RPV}} \supset \lambda'_{ijk} L_i Q_j \bar{D}_k,$$

where $i = 2$ selects the muon generation.

- The neutral-current benchmark is $p\mu^+ \rightarrow d\mu^+$, mediated by a resonant squark in the μq channel.
- The signal is selected with one reconstructed muon and at least one jet; high jet- p_T -sum thresholds suppress the DIS-like SM background.
- In favourable neutral-current channels, $\lambda'_{2j1} \sim 0.01$ - 0.02 can be tested for $m_{\tilde{q}} \sim 1$ - 3 TeV.

[Phys. Rev. D 103 (2021) 116009.]



Color-octet muons at LH μ C: resonant μg production

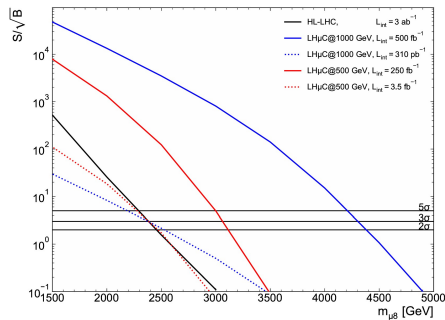


- In composite-lepton scenarios, the SM muon can have a color-octet partner μ_8 , also called a leptogluon. The effective interaction is [2506.15445 [physics.acc-ph]]

$$\mathcal{L} = \frac{1}{2\Lambda} \left\{ \bar{\ell}_8^\alpha g_s G_{\mu\nu}^\alpha \sigma^{\mu\nu} (\eta_L \ell_L + \eta_R \ell_R) + \text{h.c.} \right\}.$$

Here ℓ_8^α is the color-octet lepton, $G_{\mu\nu}^\alpha$ is the gluon field strength, Λ is the compositeness scale, and $\eta_{L,R}$ are chiral factors.

- The benchmark assumes $\Lambda = m_{\mu_8}$ and targets the resonant single-production channel $\mu g \rightarrow \mu_8 \rightarrow \mu g$.
- Signal topology: $\mu p \rightarrow \mu_8 X \rightarrow \mu j X$, with one high- p_T muon and one high- p_T jet.
- Dominant background: $\mu p \rightarrow \mu j X$, from SM neutral-current DIS-like scattering.
- The key advantage is the direct μg initial state; pp searches mainly rely on QCD pair production.



Representative 5 σ reach

Setup	m_{μ_8} reach
HL-LHC, 1 ab^{-1}	~ 2.3 TeV
LH μ C, $E_\mu = 500$ GeV, 250 fb^{-1}	~ 3.0 TeV
LH μ C, $E_\mu = 1$ TeV, 500 fb^{-1}	~ 4.2 TeV

Toward FCC- μh : final outlook and take-home points



- The long-term perspective goes well beyond the 5.3 TeV baseline.
- The FCC-hh era is explicitly connected to a future high-energy μp option, turning the current LH μ C concept into a stepping stone toward even more powerful HET and BSM programmes.
- In particular, a future FCC- μh machine, with representative $\sqrt{s_{\mu p}} \sim 15$ and 30 TeV configurations would extend:
 - Higgs statistics and coupling reach,
 - top and rare-top production,
 - high-mass electroweak and $\gamma\gamma$ studies,
 - and resonant BSM sensitivity.

HET@LH μ C

- (1) LH μ C is a serious HET factory;
- (2) the top sector is one of its clearest near-term strengths;
- (3) Higgs, EW, EFT, and $\gamma\gamma$ studies are all strongly motivated;
- (4) the concept also provides a natural bridge toward FCC-era μp physics.

Summary and Conclusions



QCD & DIS frontier

Multi-TeV μp scattering extends the accessible (x, Q^2) plane well beyond HERA and the EIC/LHeC, strengthening the PDF, small- x , and α_s programme.

Higgs–EW precision

Large CC/NC Higgs samples enable concrete coupling studies; the $h \rightarrow b\bar{b}$ benchmark reaches sub-percent sensitivity to g_{hbb} , while EW/SMEFT observables gain a larger high- Q^2 lever arm.

Top and rare-top physics

The top sector gives the clearest rate gain: already the 500 GeV LH μ C benchmark enhances $t\bar{t}$ production by ~ 32 relative to the 50 GeV LHeC case, motivating precision top and FCNC studies.

$\gamma\gamma$ and BSM portals

Photon-induced channels provide high-statistics $\tau^+\tau^-$ samples and high-mass $WW, ZZ, t\bar{t}$ probes, while direct μq and μg portals target RPV squarks, color-octet muons, and other muon-coupled BSM states.

Looking ahead: LH μ C is not simply an energy upgrade of the LHeC; it opens a multi-TeV lepton-hadron regime where QCD, Higgs, electroweak, top, photon-induced, and several BSM studies become quantitatively competitive.



The HELP4CERN collaboration



HELP4CERN collaboration focuses on the physics case for high-energy, high-luminosity lepton-hadron colliders at CERN.

- assess the physics potential of future lepton-hadron collider options,
- develop benchmark studies for Higgs, electroweak, top, QCD, and BSM physics,
- provide a common framework for phenomenology, detector studies, and future projections,
- new contributions and physics studies are very welcome,

Contact: help-4-cern@cern.ch

Visit: <https://help4cern.web.cern.ch>

White Paper on High Energy, High Luminosity Lepton-Hadron Colliders at CERN

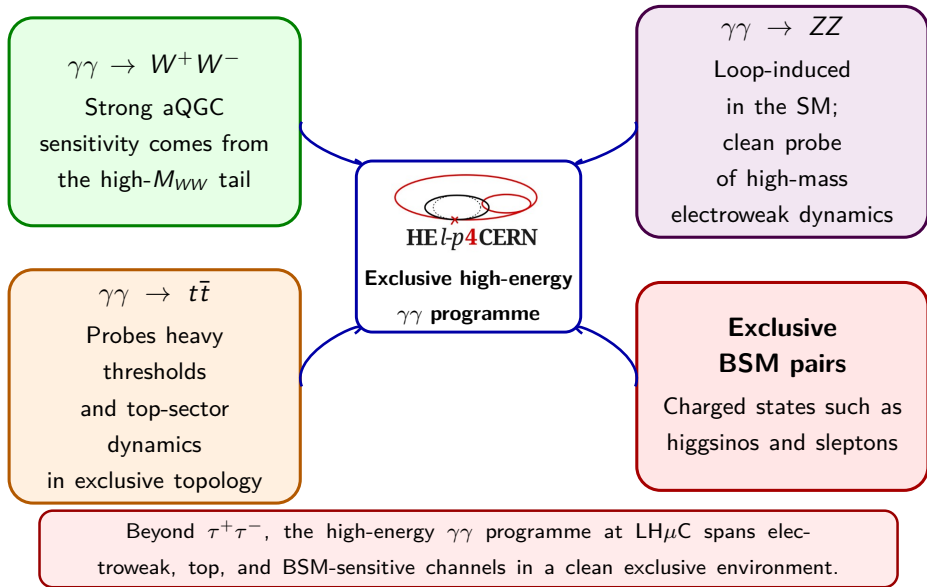


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Next steps for HELP4CERN:

- turn the current benchmark studies into a coherent white-paper assessment of high-energy lepton-hadron options at CERN by the end of 2026;
- foster a worldwide study effort, in close coordination with the LHeC Study Group, aimed at preparing a dedicated **HELP4CERN proposal** by 2028.

Exclusive $\gamma\gamma$ channels: electroweak, top, and BSM reach

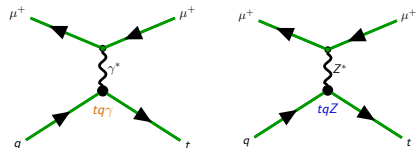


Top-FCNC probes at LH μ C: $tq\gamma$ and tqZ



- In the Standard Model, top-FCNC transitions are forbidden at tree level and remain highly suppressed at loop level by the GIM mechanism; any observable signal would therefore be a clean indication of physics beyond the Standard Model.
- The key rare-top vertices are $tq\gamma$ and tqZ with $q = u, c$, naturally parameterized by anomalous couplings or SMEFT dimension-6 operators.
- At LH μ C, the large top sample, harder kinematics, and characteristic μp topology make anomalous single-top and associated production natural benchmarks for converting top statistics into BSM reach.

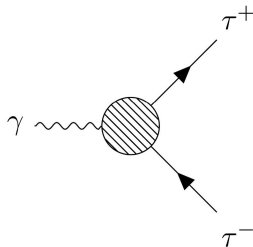
Top-FCNC at LH μ C



Representative process: $\mu^+ q \rightarrow \mu^+ t$, with $q = u, c$. Neutral-current exchange probes anomalous $tq\gamma$ and tqZ couplings.

Improved sensitivity relative to the LHeC is expected at LH μ C, but quantitative limits remain channel-dependent and require realistic event selections, detector modelling, and systematic-uncertainty estimates.

Current constraints on a_τ , d_τ



Benchmark topology

Photon-induced

$$\gamma\gamma \rightarrow \tau^+\tau^-$$

directly probes the $\gamma\tau\tau$ vertex. It is therefore a clean channel for anomalous magnetic and electric dipole moments, a_τ and d_τ .

Current 95% CL constraints on a_τ

Experiment	Channel	Dataset	a_τ interval
DELPHI	$e^+e^- \rightarrow e^+e^-\tau^+\tau^-$	650 pb ⁻¹	[-0.052, 0.013]
ATLAS 2022	PbPb UPC	1.44 nb ⁻¹	[-0.057, 0.024]
CMS 2022	PbPb UPC	1.34 nb ⁻¹	[-0.030, 0.017]
CMS 2024	pp exclusive $\gamma\gamma \rightarrow \tau\tau$	138 fb ⁻¹	[-0.0022, 0.0041]
ATLAS 2025	pp high-mass $\tau\tau$	140 fb ⁻¹	[-0.0024, 0.0047]

- The SM expectation is

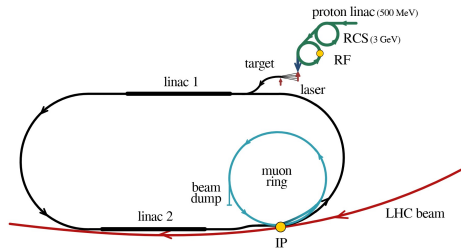
$$a_\tau^{\text{SM}} \simeq 117721 \times 10^{-8} \simeq 1.18 \times 10^{-3}.$$

- CMS 2024 also reports

$$a_\tau = 0.0009_{-0.0031}^{+0.0032}, \quad |d_\tau| < 2.9 \times 10^{-17} \text{ e cm} \quad (95\% \text{ CL}).$$

- Present LHC constraints are therefore already at the few- 10^{-3} level for a_τ . The LH μ C motivation is a much larger high-mass exclusive $\tau^+\tau^-$ sample.

Technical concept behind the LH μ C proposal



Conceptual layout of the proposed LH μ C configuration.

- The LH μ C proposal aims to re-use key elements of the LHeC infrastructure to make high-energy, high-luminosity μ^+p collisions.
- The basic ingredients are:
 - an ultra-cold positive-muon source,
 - acceleration / booster stages for the μ^+ beam,
 - a dedicated μ^+ storage ring.

See talk by Krzysztof Piotrkowski

References

- (1) μ TRISTAN: Y. Hamada *et al.*, *Prog. Theor. Exp. Phys.* **2022** (2022) 053B02.
- (2) μ LHC: Antimuon Ring and High Luminosity-LHC-Based μ^+p Collider: D. Akturk *et al.*, *Prog. Theor. Exp. Phys.* **2026** (2026) 033C04.
- (3) HELP4CERN White Paper (in preparation)